

Capstone Project Proposal (Data Analysis Pathway)

Project Title: *Roadway Risk for Pedestrians and Cyclists in Louisville, KY (2018 - 2022)*

Proposed By: Josh Groft

GitHub Repository: https://github.com/JishGorft/Roadway_Risk

Project Overview

Using public open data from various government offices and safety initiatives, this project aims to assess the general risk to pedestrians and cyclists in Louisville, Kentucky. While these data exist for public access, no resource is provided that explicitly combines the critical “high injury network” data created as part of the Vision Zero project, which aims to eliminate roadway deaths in the city by 2050, with crosswalk, bike path, or bus stop data. The goal is to determine how much overlap exists between the most dangerous stretches of road, in terms of roadway deaths and injuries, and common pathways for pedestrians and non-motor vehicles.

Data Sources

In the lists of relevant fields below, “location” may indicate latitude and longitude coordinates, roadway mile markers, intersections, landmarks, or other indicative data.

Data Set	Source	Relevant Fields	URL
High Injury Network	Louisville Metro Open Data	location, fatal and serious injury accident counts, priority ranking (seriousness of crashes ranked), comprehensive crash cost (“EPDO”)	https://data.louisvilleky.gov/datasets/LOJIC::louisville-metro-ky-high-injury-network-1/about
Road Context Classifications	Louisville Metro Open Data	location, speed limit, type of roadway (i.e. highway, avenue, lane)	https://data.louisvilleky.gov/datasets/LOJIC::louisville-metro-ky-road-context-classifications/about
Midblock Crossings	Louisville Metro Open Data	location, aid type (i.e. rumble strip, speed hump), crossing type (i.e. school, pedestrian), attraction (landmark or reason for crosswalk if not intersection)	https://data.louisvilleky.gov/datasets/LOJIC::jefferson-county-ky-midblock-crossings/about
Jefferson County KY Bikeways	Louisville Metro Open Data	location, bikeway type (shared lane, dedicated lane, trail)	https://data.louisvilleky.gov/datasets/LOJIC::jefferson-county-ky-bikeways/about
TARC Bus Stops	Louisville Metro Open Data	location	https://data.louisvilleky.gov/datasets/LOJIC::louisville-metro-area-ky-tarc-bus-stops/about

Research Objectives

- **Primary Question(s):** How much of Louisville's identified high injury network is comprised of high-pedestrian or high-cyclist areas?
- **Secondary/Exploratory Questions:** What safety measures are already deployed in such areas? What are the speed limits in such areas?

Louisville is a Vision Zero city, which aims to eliminate roadway fatalities in the city by 2050. The high injury network database was created to assist in this initiative, but no data on what steps have been taken so far is available. There also appears to be no existing analysis of whether or how high-pedestrian and high-cyclist corridors intersect with the high injury network.

Data Preparation Approach

- Align data sets on available location data.
- Coming from government source, data is already clean and largely standardized. This is not true of location data in all tables.
- Ensure consistent location data available in all tables. Some tables include latitude and longitude, while others include intersections, or the coordinate geometry of specific stretches of road.
 - High Injury Network, Road Context Classifications, and Midblock Crossings contain consistent roadway coordinate keys (RWCOMPKEY) on which mapping will, in theory, be simple.
 - Bikeways and TARC Bus Stops tables include only geographic coordinates.
- Devise a method to infer RWCOMPKEY from geographic coordinates, or create new keys in all tables by merging coordinate data across tables (if possible).
- Combine Road Context Classifications and High Injury Network.
- Create one or more composite indicators of risk assessment:
 - *Pedestrian Risk Index* (crosswalk + bus stop + high injury network)
 - *Cyclist Risk Index* (bikeways + high injury network)
- Store data in a SQLite database with tables:
 - road_details (location, speed_limit, road_type, injury_rank)
 - pedestrians (location, bus_stop, safety_measure, attraction, crosswalk_type)
 - bikeways (location, bikeway_type)

Current Status

- Data acquired and inspected.
- Early cleaning or standardization steps have been taken.
- Prototype exploratory plots have been created.
- Most data sets have been merged via pandas successfully.

Deliverables (Remaining Work)

Required Tasks (must be completed):

- *Finalize SQLite schema and load tables.*
- *Write and test three functions.*
- *Generate required visualizations: likely examples include several maps showing high injury network with coinciding pedestrian and bikeway areas. Plot risk assessments (composite indicators) by neighborhood, corridor, or other distinguishing location feature.*
- *Complete GitHub repository with data, project notebook, README.md, data dictionary, and requirements.txt.*

Stretch Goals (optional, if time allows):

Build interactive map plot to allow for user to see stacked layers, i.e. high injury network heat-mapped by risk, bus stops, crosswalks, and bikeways.

Project Timeline

- Phase 1: Acquire data, exploratory data analysis, and data consolidation.
 - Download data sets,
 - Assess available fields,
 - Trim and combine datasets in geopandas.
- Phase 2: Set up database, build functions, first draft visualizations.
 - Establish tables (outlined above),
 - Create composite risk indicators,
 - Plot maps for further analysis (potentially honing in on specific areas of Jefferson County).
- Phase 3: Deeper analysis, refine visuals, draft report.
 - Analyze maps with focus on composite indicators.
 - Delimit existing or create new map plots of areas highlighted by composite indicators.
- Phase 4: Finalize deliverables, clean repository, record presentation.
 - Polish repository structure (README, requirements.txt, ERD diagram, schema file).
 - Finalize notebooks with analysis narrative and visuals.
 - Submit complete GitHub repository.

Additional Considerations

Geodata may prove challenging to work with, particularly in defining shared keys across tables. Some geodata describes singular points, while other geodata describes lines or polygons. Database tables may need to be built considering geoshapes as entities, with commonalities (i.e. points and lines fall within certain quadrants) as distinct fields.

High Injury data set also includes data related to the federal [Justice40 Initiative](#). While this program has been [defunct since 2025](#), it may prove useful to include these elements in analysis, i.e. Is the percentage of crosswalks and bike lanes in high injury areas disproportionately represented in underprivileged neighborhoods?